

## Executive Summary

The HP lattice model is a classic NP-hard protein folding problem <sup>1</sup>. Exact **constraint programming (CP)** and integer-programming methods (e.g. CPSP-tools <sup>2</sup>, ILP models <sup>3</sup>) already solve 3D HP instances optimally for moderate sizes. Likewise, **deep reinforcement learning (RL)** (notably DQN with LSTM variants) has recently been applied: Yang *et al.* (2022) solved 2D HP sequences up to length 50 <sup>4</sup>, and Espitia *et al.* (2024) solved 3D HP (up to ~58 residues) with a hybrid DQN/LSTM-Attention approach <sup>5</sup>. In contrast, **no prior work explicitly applies Google OR-Tools CP-SAT to the 3D HP model**, and no study directly juxtaposes a CP solver and a DRL agent on identical 3D HP instances. The third claim — that both methods plateau at similar (suboptimal) energies on long sequences — appears to be an **original observation** not documented in the literature.

In summary:

- **Claim 1 (CP-SAT on 3D HP):** CP methods for 3D HP are well known (CPSP <sup>2</sup>, Backofen & Will <sup>6</sup>); CP-SAT *specifically* is new to this domain. We find this **incremental**, not fundamentally novel.
- **Claim 2 (CP vs RL head-to-head):** No prior study directly compares CP (or ILP) vs DRL for HP folding. This direct empirical comparison is **novel in execution** but conceptually incremental (the methods themselves are known).
- **Claim 3 (Converging to similar suboptimal energies):** This convergence has not been reported elsewhere and is a **new empirical insight**, though its significance may be modest.

**Recommendation:** As a 3rd-year project, this work is well done, but on its own it **lacks a sufficiently novel algorithmic contribution**. It would **not survive** stringent peer review at top ML/AI, CP/OR or computational biology venues, though it might be acceptable at a specialized workshop or student conference after focused improvements. (See table and discussion below for details.)

```
timeline
  title: Timeline of Key HP Folding Research
  1998 : Berger & Leighton prove HP (3D-cubic) folding NP-complete 1.
  2006 : Backofen & Will (Constraints'06) use CP to find optimal 3D HP
  folds 6.
  2008 : Mann *et al.* introduce CPSP-tools: first high-throughput exact 3D
  HP solver 2.
  2015 : Nunes *et al.* (DAM'15) model 3D-HP with side chains as an ILP,
  solved with CPLEX 3.
  2022 : Yang *et al.* (Physica A'22) use DQN+LSTM on 2D HP (length 20-50),
  finding best-known energies 4.
  2024 : Espitia *et al.* (arXiv'24) apply reservoir-DQN and LSTM-Attn to
  **3D** HP (length ≤58), matching known optima 5.
```

### Claim 1: CP-SAT applied to 3D HP

**What's known:** 3D HP folding has been tackled by exact CP and integer/ILP methods. For example, Backofen & Will (2006) presented a CP formulation for the cubic lattice HP model <sup>6</sup>, and the CPSP-

tools (Mann *et al.* 2008) implement sophisticated CP algorithms to enumerate all optimal structures in cubic and FCC lattices <sup>2</sup>. An ILP model (Nunes *et al.* 2015) solves 3D HP with side-chains via CPLEX <sup>3</sup>. These methods produce provably optimal energies.

**Incremental novelty:** Google OR-Tools' CP-SAT solver is a modern constraint solver, but using it on HP folding is essentially an **engineering application** of known CP techniques. We find *no prior citation* of CP-SAT being used for HP folding. Thus the project's first claim is that *it applies CP-SAT to 3D HP*. This is new in implementation but conceptually incremental: it reuses established CP ideas with a new solver. There is no new modeling insight beyond choosing OR-Tools, so **publishable novelty is minimal**. In other words, CP-SAT solves HP similarly to earlier CP libraries (Gecode, ILOG) <sup>2</sup> <sup>6</sup>, so the solver choice alone is not scientifically novel.

**Evidence:** Literature on lattice HP (e.g. CPSP-tools <sup>2</sup>) emphasizes CP without naming OR-Tools. We saw nothing indicating prior CP-SAT use on HP, only mentions of "constraint programming" in general <sup>6</sup> <sup>2</sup>. Therefore:

- **Recommendation: Incremental.** CP on 3D HP is well studied, and OR-Tools CP-SAT is just a new implementation.

## Claim 2: Direct CP-SAT vs RL comparison

**What's known:** Deep RL for HP is quite new. Yang *et al.* (2022) applied DQN with LSTM to the 2D HP model, finding best-known folds for sequences up to length 50 <sup>4</sup>. Espitia *et al.* (2024) extended DRL to **3D HP**, using reservoir/DQN for short chains and an LSTM with multi-head attention for longer chains, also reaching optimal energies <sup>5</sup>. However, these RL studies **do not compare directly to CP/exact solvers**. Likewise, CP/ILP methods aim for provable optimality (CPSP-tools <sup>2</sup>, ILP <sup>3</sup>), but have not been benchmarked against DRL methods on the same inputs in the literature. We found no prior "head-to-head" empirical comparison of CP vs RL on any HP lattice problem.

**Incremental novelty:** The project's second claim is an empirical comparison: "*CP-SAT vs our RL agent (LSTM-A / DQN) on the same 3D HP sequences.*" While comparing methods is useful, it is not conceptually new. It reveals relative performance (e.g. runtime, solution quality), but the techniques themselves are borrowed. Such comparative experiments are **novel data** in this domain, but represent an incremental step.

Reviewers would note that both CP (exact) and DRL (heuristic) are known tools. Demonstrating which yields lower energy under time limits could be informative, but without a new algorithmic twist it is not a breakthrough contribution. The question "Which is better, CP or RL, on HP folding?" is unanswered, so doing so is new. However, as a "claim", it's largely a straightforward experiment rather than a novel method.

**Evidence:** No journal or conference papers appear to perform this exact CP-vs-RL comparison. Most HP folding literature compares *stochastic heuristics vs CP* (e.g. ant colony <sup>7</sup>, GA <sup>8</sup>), but not RL. The nearest is Espitia's work <sup>5</sup>, which cites prior (Yang 2022) and focuses on RL architectures, not CP. Thus:

- **Recommendation: Incremental/Near-novel.** The comparison itself hasn't been done, so the empirical finding is new. But the lack of a methodological innovation beyond running two existing methods makes it an incremental contribution, not a fundamentally publishable advance.

## Claim 3: Convergence to similar suboptimal energies

**What's known:** Neither RL-HP nor CP-HP papers discuss the specific phenomenon that *both methods plateau at the same suboptimal energy as sequences grow longer*. Yang et al. and Espitia et al. report that their RL agents reach the best-known energies for tested sequences <sup>4</sup> <sup>5</sup>, but they do not contrast this with CP performance or note a convergence issue. CP approaches (CPSP-tools <sup>2</sup>) simply report optimal solutions (or fail after exhaustive search). We found no mention in the literature of a “common limitation” where even powerful solvers/learners hit the same energy ceiling on longer HP sequences.

**Novel insight:** This empirical observation could be useful (suggesting intrinsic hardness for long HP folding). Since it isn't documented elsewhere, it counts as an original contribution. However, it arises from the project's experiments rather than a new theory. It's possible it reflects known hardness (HP NP-completeness <sup>1</sup>) manifesting in practice. As a standalone claim, it's an interesting insight but not a dramatic surprise.

- **Recommendation: Novel (but limited significance).** Reporting this phenomenon is a new finding. It may attract curiosity but likely wouldn't alone justify a high-impact publication; it would need deeper analysis or explanation.

## Summary of Prior Work

| Paper (Year)                              | Method                               | Variant (Dim, model)   | Sequences/ Benchmarks                          | Main Result                                                                 |
|-------------------------------------------|--------------------------------------|------------------------|------------------------------------------------|-----------------------------------------------------------------------------|
| Berger & Leighton (1998) <sup>1</sup>     | Complexity theory                    | 3D HP (cubic)          | —                                              | HP model folding NP-complete (3D) <sup>1</sup> .                            |
| Backofen & Will (2006) <sup>6</sup>       | Constraint Programming (Gecode)      | 3D HP (cubic FCC)      | Standard HP sequences (e.g. length $\leq 48$ ) | Provably optimal structures found in 3D HP.                                 |
| Mann <i>et al.</i> (2008) <sup>2</sup>    | CPSP-tools (CP + custom algorithms)  | 3D HP (cubic & FCC)    | Large collections of short sequences           | First <b>exact</b> high-throughput solver; reports all optima and energies. |
| Nunes <i>et al.</i> (2015) <sup>3</sup>   | Integer Programming (ILP, CPLEX)     | 3D HP side-chain       | 3 sets of literature-based instances           | Optimal contacts found; new bounds for side-chain HP.                       |
| Yang <i>et al.</i> (2022) <sup>4</sup>    | Deep RL (DQN with LSTM)              | 2D HP (square lattice) | 8 benchmark sequences (n=20–50)                | Found best-known energies; LSTM boosts learning.                            |
| Espitia <i>et al.</i> (2024) <sup>5</sup> | Deep RL (Reservoir + DQN, LSTM-Attn) | 3D HP (simple cubic)   | 3D conformations (n<36 and n $\leq$ 58)        | Achieve optimal energies; faster convergence with hybrid model.             |

| Paper (Year)            | Method                                   | Variant (Dim, model)         | Sequences/ Benchmarks           | Main Result                                                      |
|-------------------------|------------------------------------------|------------------------------|---------------------------------|------------------------------------------------------------------|
| <b>This work (2026)</b> | CP (OR-Tools CP-SAT) vs. RL (LSTM-A/DQN) | 3D HP (cubic, backbone-only) | Sequences "3d1"–"3d8" (n~20–58) | Both methods find similar suboptimal energies at longer lengths. |

Each above paper (and this project) focuses on HP model variants on 2D or 3D lattices. Earlier CP/ILP works guarantee optimality <sup>6</sup> <sup>3</sup>, while the RL works deliver best-known (not guaranteed optimal) solutions <sup>4</sup> <sup>5</sup>. **No prior work** combines a CP solver with a DRL agent on the same benchmark for HP, nor notes their solution energies converging.

## Novelty Assessment by Claim

- **(1) CP-SAT on 3D HP: Incremental.** CP/optimization has long been applied to HP folding <sup>6</sup> <sup>2</sup>. Using Google's CP-SAT is a new twist, but it is simply a new solver rather than a new algorithmic idea. We did not find any prior literature on OR-Tools specifically for HP, so the project is the first to report it, but this is a minor novelty. For peer review, "we solved HP with OR-Tools" would be judged as an implementation note, not as a major advance.
- **(2) CP vs RL comparison: Incremental / Partly novel.** Directly measuring CP-SAT against an RL agent on HP is not previously reported (so it's novel experimental data), but conceptually it's a straightforward empirical study. Journals/conferences typically expect either a new method or a surprising insight; a plain head-to-head does not suffice unless it reveals something fundamentally interesting. Here it mainly confirms that two known techniques have comparable difficulties on harder instances. This is a useful observation but would be seen as incremental: it validates that *neither approach dramatically beats the other* on these tests. Reviewers will likely see the comparison as descriptive, not prescriptive.
- **(3) Converging suboptimal energies: Novel (empirical insight).** No paper seems to have noticed that CP-SAT and RL both stall at the same energy level on long chains. If true, it suggests an intrinsic limit of current methods. While an interesting observation, it is not a theoretical result and on its own might not merit a full publication. However, it is a valid new claim about HP folding heuristics.

In all cases, the claims are modest extensions of existing knowledge. There are no breakthrough algorithms or proofs. The contributions are primarily experimental observations. As such:

- Overall, the work **lacks the level of novelty** typically required for strong peer-reviewed venues in ML/AI, CP/OR or computational biology <sup>2</sup> <sup>5</sup>. It is at best a "nice study" with some minor new data.

## Venue Suitability and Conclusion

**Top-Tier ML/AI (e.g. NeurIPS, ICML/AAAI full papers):** These venues demand substantial algorithmic novelty or empirical breakthroughs. This project's contributions would be deemed **insufficiently novel**. A main conference submission would almost certainly be **rejected**.

**Specialized Workshops (NeurIPS ML4Bio, CP workshops, student tracks):** With revisions, it might be acceptable as a workshop or poster. The empirical angle (CP vs RL) could interest niche audiences, but the work needs polishing (stronger analysis, clearer insight, more datasets).

**Constraint Programming/Operations Research (e.g. CP conference):** Reviewers would note that CP applications to HP are well-known <sup>6</sup> <sup>2</sup>. Without new CP methodology (just using CP-SAT) or a deeper theoretical contribution, acceptance is unlikely except as a short demonstration or contest entry.

**Computational Biology/Bioinformatics (e.g. RECOMB, BMC Bioinformatics):** These audiences value biological relevance and clear performance gains. The HP model is very abstract, and this work does not address real proteins or novel biology. It is more of an optimization/ML exercise. Thus, it would **not meet novelty criteria** for a journal like Bioinformatics or PLOS Comp Bio. Possibly a short conference report could pass if framed as “application note”.

**Conclusion:** As currently formulated, the manuscript **would not survive** rigorous peer review in its field. Its strongest chance might be as a workshop paper (ML or CP) after emphasizing the novel observations and situating them against the cited HP literature <sup>6</sup> <sup>5</sup>. However, to become genuinely publishable, the authors should (a) highlight any unique insight beyond reproducing known results, and (b) consider methodological extensions (e.g. hybrid CP+RL strategies) that go beyond mere comparison.

**Final recommendation (by claim):**

- CP-SAT on 3D HP – *Incremental* (already-known domain, new solver).
- CP vs RL comparison – *Incremental/Novel* (novel comparison, but expected outcome).
- Similar energy convergence – *Novel insight* (not previously reported).

**“This would not survive peer review at major venues (ML/AI, CP, Bioinformatics) due to limited novelty and absence of a clear new method, although it could potentially fit a smaller workshop or student track.”** <sup>2</sup> <sup>5</sup>

---

<sup>1</sup> [computingbiology.github.io](https://computingbiology.github.io)

<https://computingbiology.github.io/docs/berger1998.pdf>

<sup>2</sup> <sup>7</sup> (PDF) CPSP tools-Exact and Complete Algorithms for High-throughput 3D Lattice Protein Studies  
[https://www.researchgate.net/publication/5390121\\_CPSP\\_tools-Exact\\_and\\_Complete\\_Algorithms\\_for\\_High-throughput\\_3D\\_Lattice\\_Protein\\_Studies](https://www.researchgate.net/publication/5390121_CPSP_tools-Exact_and_Complete_Algorithms_for_High-throughput_3D_Lattice_Protein_Studies)

<sup>3</sup> (PDF) An integer programming model for protein structure prediction using the 3D-HP side chain model  
[https://www.researchgate.net/publication/282627360\\_An\\_integer\\_programming\\_model\\_for\\_protein\\_structure\\_prediction\\_using\\_the\\_3D-HP\\_side\\_chain\\_model](https://www.researchgate.net/publication/282627360_An_integer_programming_model_for_protein_structure_prediction_using_the_3D-HP_side_chain_model)

<sup>4</sup> [2211.14939] Applying Deep Reinforcement Learning to the HP Model for Protein Structure Prediction  
<https://arxiv.org/abs/2211.14939>

<sup>5</sup> [2412.20329] Protein Structure Prediction in the 3D HP Model Using Deep Reinforcement Learning  
<https://arxiv.org/abs/2412.20329>

<sup>6</sup> HPstruct - Optimal structure prediction  
<http://cpsp.informatik.uni-freiburg.de/HPstruct/Input.jsp;jsessionid=2279709427691A953C030CD20C6020B6>

8 Firas Khatib's research works | University of Massachusetts Dartmouth and other places  
<https://www.researchgate.net/scientific-contributions/Firas-Khatib-2082719244>